

Modelling & Simulation: Assignment 5

2-D robot arm with SEAs

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Introduction

You are going to model a planar robot arm with two degrees of freedom: two 2-D rigid bodies with two joints. The two joints should be actuated by series-elastic actuators (SEAs). The robot arm is going to execute some simple task, namely, moving its end-effector (endpoint of the second link) back and forth between two positions. To make your system safe for human-robot interaction, we want the robot arm to be slightly compliant, that is to say, you must use a form of impedance control. You must show this compliance by demonstrating that your robot arm “gives way” when you apply an external force on the end-effector (the tip of the arm).

Systems like robotic arms, with rigid bodies connected end-to-end by joints, are commonly referred to as *serial kinematic chains*. Over the coming assignments you will study different methods of modelling such systems, both in 2-D and 3-D. In this assignment, we will focus on the energy-based Euler-Lagrange method, in 2-D. Combined with SEAs and a few assumptions, you will reproduce what is called *Spong’s model*. In this model, actuator dynamics and robot dynamics can be written separately, and the two are connected by power ports, as in Fig. 1.

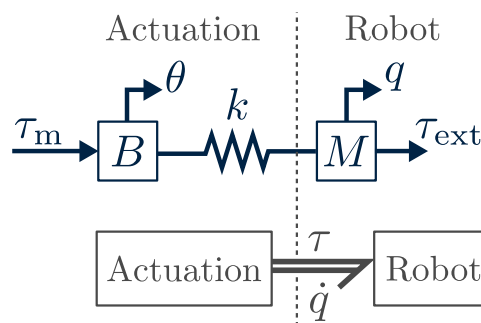


Figure 1: Series-elastic actuator driving a robot.

Suggested material:

- M. W. Spong, “Control of flexible joint robots: A survey”, 1990. (Sections 1–3 on modelling, Eq. (8)-(9)).
- Slide decks on Lagrangian mechanics on Canvas (I found [Multibodies.pdf](#) helpful!).
- G. A. Pratt and M. M. Williamson, “Series elastic actuators”, IEEE/RSJ International Conference on Intelligent Robots and Systems, vol. 1, pp. 399–406, 1995.
- W. Roozing, J. Malzahn, N. Kashiri, D. G. Caldwell, and N. G. Tsagarakis, “On the Stiffness Selection for Torque-Controlled Series-Elastic Actuators”, IEEE Robotics and Automation Letters, vol. 2, no. 4, pp. 2255–2262, doi: 10.1109/LRA.2017.2726141, 2017.

Assignment

Model a 2-DoF planar robot arm using the Euler-Lagrange method. Actuate it by series-elastic actuators, achieving Spong's model with actuator submodels connected to your robot model via power ports. Implement some form of impedance control and show proper functioning in a simulation experiment.

You may make the assignment as simple or as difficult as you like... Some ideas below.

- Will you include centrifugal/coriolis terms, or are they negligible at the speeds you are considering?
- Will you implement a joint-level impedance controller of the form $\tau = K(q_{\text{ref}} - q) - b\dot{q}$? Or a workspace (Cartesian) impedance control law ($F = K(x_{\text{ref}} - x) - bv$), properly mapped to joint torques (hint: Jacobian)?
- Does your robot arm move in a horizontal or vertical plane? If vertical, are you going to compensate gravity in your controller?
- What kind of external force are you going to put on the end-effector? What stiffness/compliance should your arm show? When do you think it can be considered "safe"? (What is the size and weight of your links?)
- If you have time: Will you include the electrical motor dynamics, current control, discrete-time control and sampling, etc?

A few hints:

- On Canvas there is an example of an equation submodel that implements a pendulum using its differential equations, and providing a power port $\langle \dot{q}, \tau \rangle$ to the outside. You can use this to develop your 2-DoF arm model.
- As always, consider simplifying initially, and perform **explicit verification steps**. For example,
 1. Leave the arm unactuated to see if it swings under gravity as you would expect,
 2. Develop and test your actuators separately, with fixed output or a simple mass as load (this also allows to parallelise the work within the group!),
 3. Try to get your impedance control working first by directly applying torques instead of using your actuator models,
 4. This is simulation; you can always turn gravity off during testing.
 5. Try to break your own model; what happens when you change parameters?

These are becoming complex models, and issues are becoming harder to spot in the big end results!